AE-14S 60A FOC

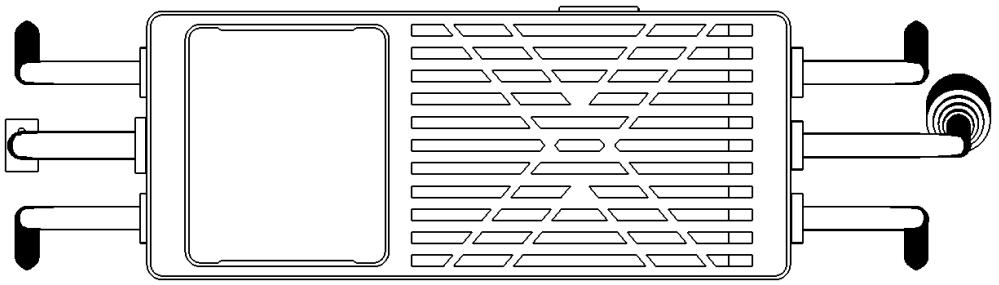
User Guide

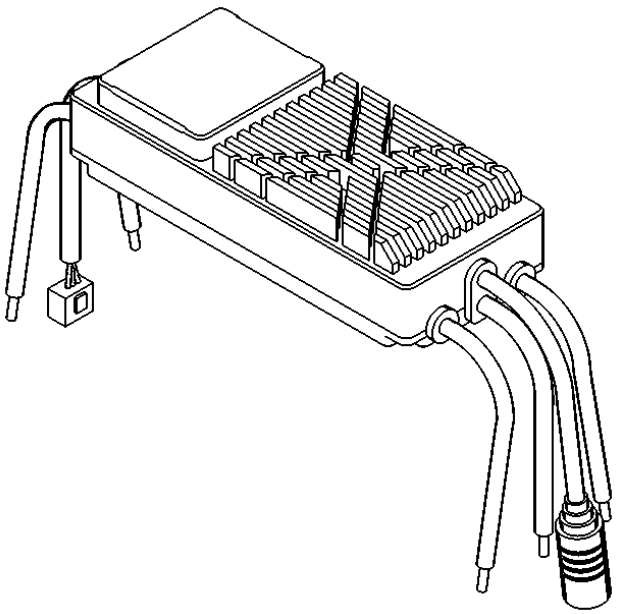


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1. Parts List

· ESC





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1. Parts List

· Wires & Cables

Wire	Type	Signals
Battery leads	14 AWG · battery end · 2 wires	BAT+ · BAT−
CAN cable	battery end · JST-GH 4P plug · 3 wires	CAN_H · CAN_L · GND
PWM input (option)	battery end · selected with can_pwm = 1 · 2 wires	PWM · PWM−
Motor leads	16 AWG · motor end · 3 wires	MOT_A · MOT_B · MOT_C
Encoder cable	motor end · round 6-pin plug (Ø 8.3 mm) · SPI · 6 wires	3V3 · GND · CS · CLK · MOSI · MISO

2. Specifications

· Hardware Specifications (1)

MCU	TI TMS320F28027F + ST STM32F103
Control	Field-oriented control with the motor's magnetic encoder
Voltage Range	6–14S LiPo (18–60 V)
Constant Current	30 A
Burst Current	60 A (10 s)
Interface	DroneCAN (CAN 2.0 · 1 Mbit/s · isolated) on a JST-GH 4P cable throttle, telemetry, settings and firmware updates
Telemetry	DroneCAN ESC status voltage, current, rpm, temperature and fault flags
Prop Parking	Holds the propeller at a stored angle at 0 % throttle
Protection	Current limited every switching cycle · heat lowers the limit, never below 50 % · short-circuit retry · voltage and throttle-loss flags
Motors	novaX AM-6212 / AM-6215 with encoder (28 poles), measured on the aircraft with auto_cal

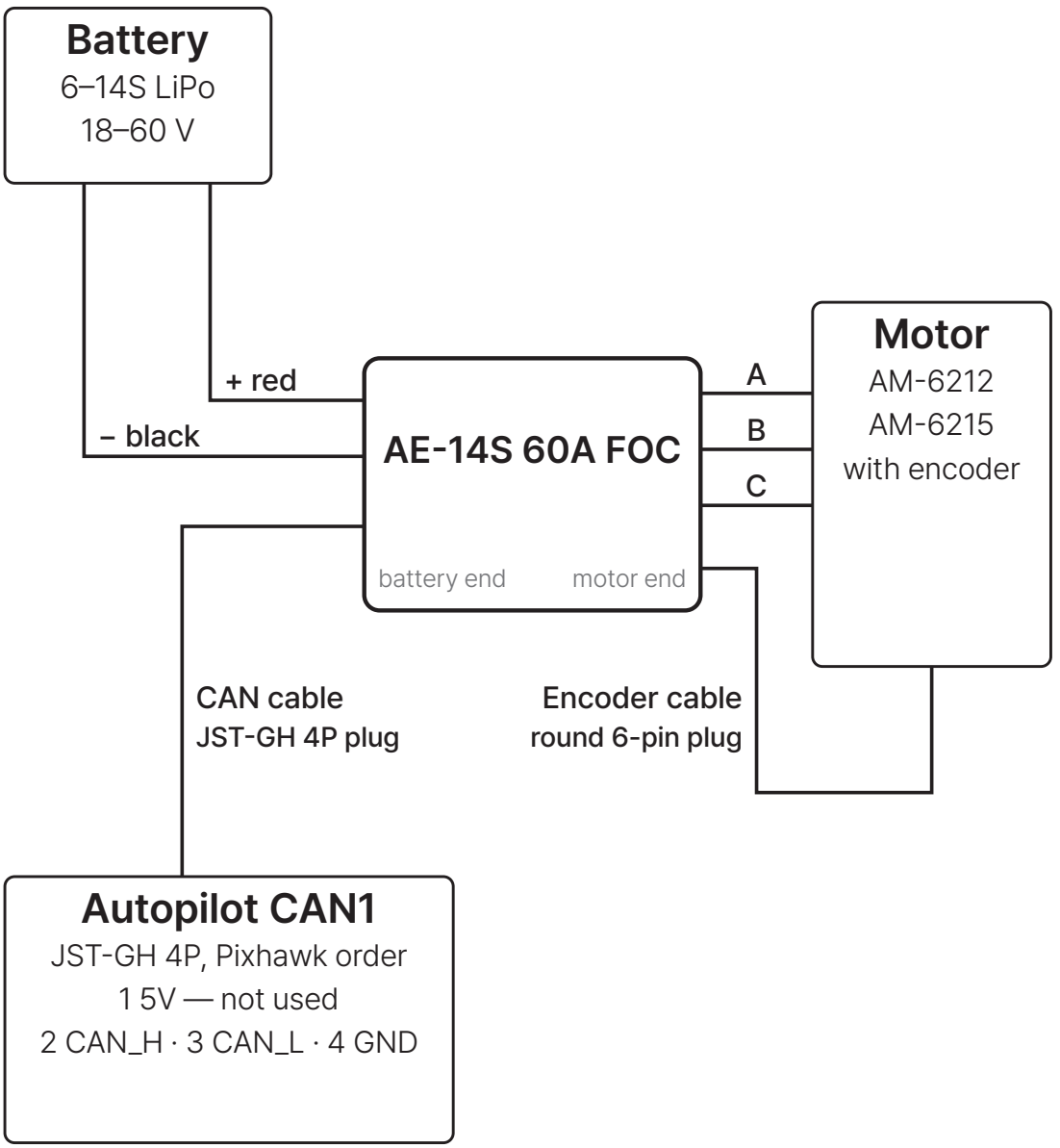
2. Specifications

· Hardware Specifications (2)

Wires	Battery 14 AWG · motor 16 AWG · CAN cable with JST-GH 4P plug · encoder cable with round 6-pin plug
Size	84 × 35 × 20 mm
Firmware	novaX FOC App V1.40 · DroneCAN 1.13, updated over DroneCAN

3. Wiring

· Connection Diagram

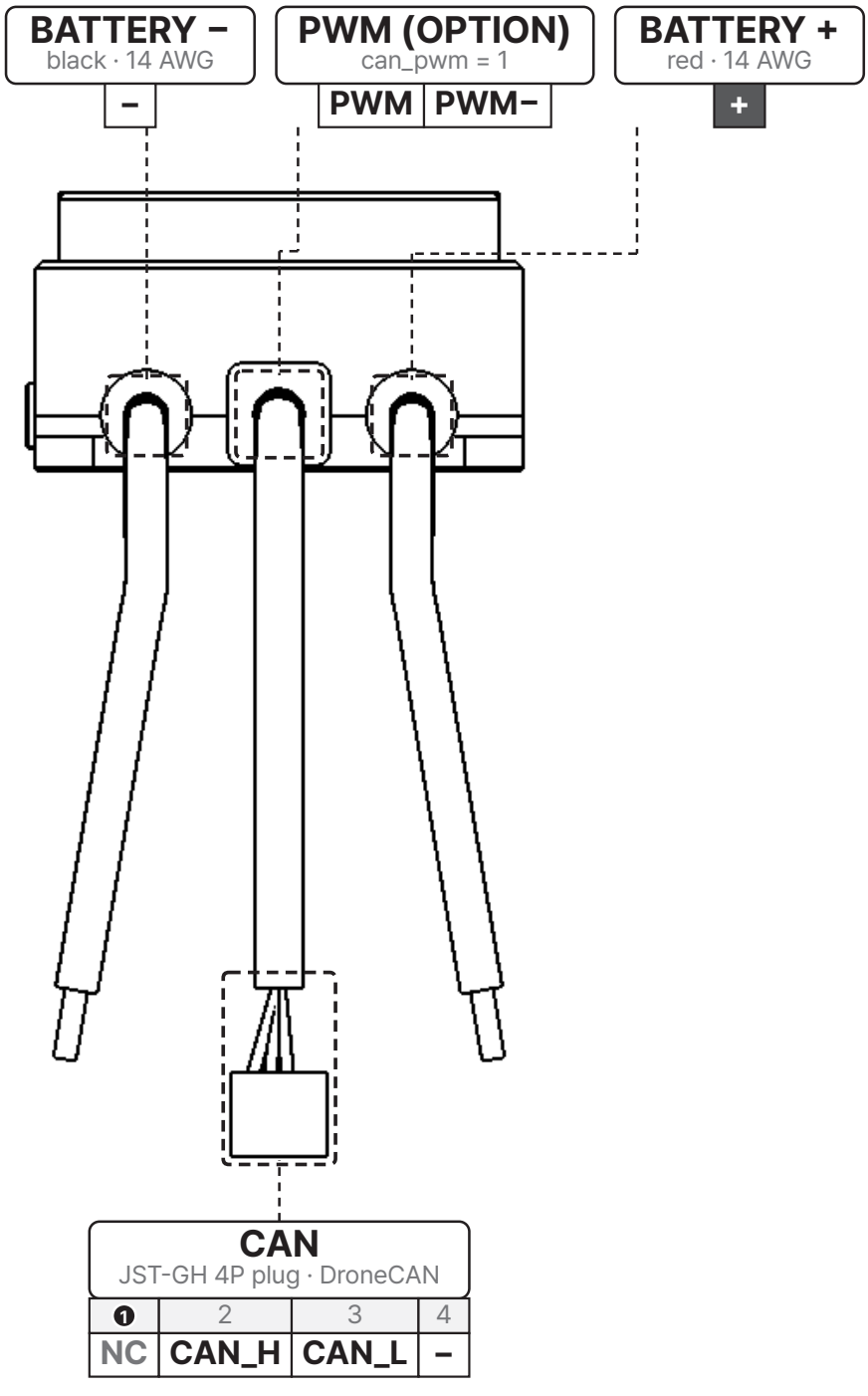


Motor leads: any order — then run auto_cal and set the direction with dir. CAN carries the throttle, telemetry, settings and firmware updates.

3. Wiring

· Battery End

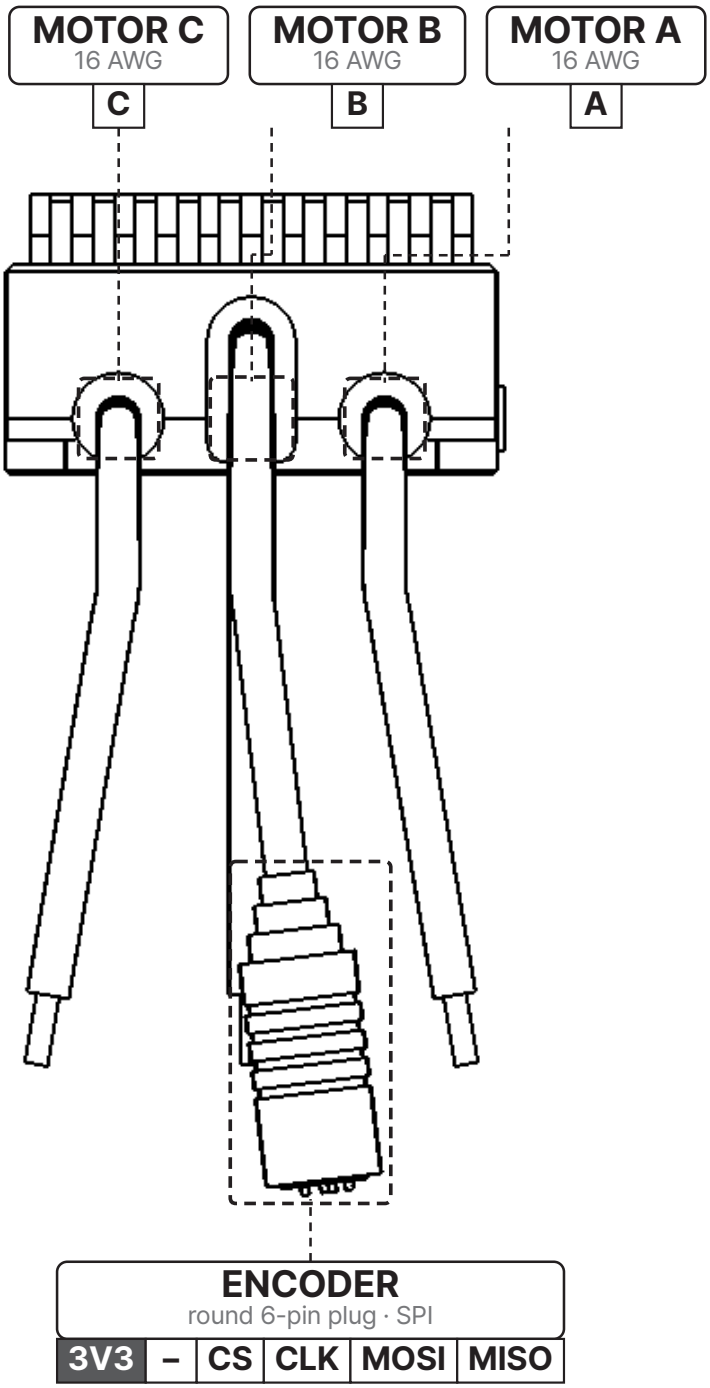
❶ = pin 1 **+** = battery **3V3** = supply **–** = GND **CS** = signal · seen from outside the case



3. Wiring

· Motor End

❶ = pin 1 **+** = battery **3V3** = supply **–** = GND **CS** = signal · seen from outside the case



4. Setup

· Step by Step (1)

Set the autopilot first (parameters below), then give every ESC its number. ESC parameters are DroneCAN parameters: in Mission Planner open SETUP → Optional Hardware → DroneCAN/UAVCAN, choose MAVlink-CAN1 and press Start. Each ESC appears as AE14S_60A_V1.40 — Menu → Parameters, change the value, Write. The ESC refuses a change while its motor turns or the autopilot is armed (the old value comes back), so disarm first.

1. Wiring

Battery leads to the power distribution (red +, black –). The three motor leads to the motor in any order. Plug the encoder cable's round 6-pin plug into the motor's encoder cable and the CAN cable's JST-GH 4P plug into the autopilot CAN1 port (its pin 1, 5 V, is not used: the ESC powers its isolated CAN side itself).

2. Autopilot

Set the ArduPilot parameters in the table and reboot. CAN_D1_UC_ESC_BM marks the outputs that are sent to DroneCAN ESCs (15 = outputs 1–4); CAN_D1_UC_ESC_OF skips that many outputs, so the first marked output becomes ESC 0 (motors on outputs 5–8: BM 240, OF 4).

3. ESC number

Give each ESC its esc_index: 0 for the first motor output, 1 for the next, and so on. Number the ESCs one at a time with only that ESC connected, or spin each with the motor test and match the rpm shown.

4. Setup

· Step by Step (2)

4. Motor measurement

Propeller off, motor cold, autopilot disarmed: write auto_cal = 1. The motor turns slowly for about 20 s while the ESC measures resistance, inductance and flux twice and calibrates the encoder. motor_meas = 1 means the values were saved; any other number tells why not (see the user guide).

5. Direction

Spin the motor with the motor test at 7–10 %. If it turns the wrong way, change dir (0 ↔ 1). Do not swap motor leads after auto_cal: the encoder calibration follows the lead order (after a swap, run auto_cal again).

6. Prop parking (optional)

Fit the propeller, turn it by hand to the parking angle and write prop_align = 1. At 0 % throttle the ESC holds the propeller there. Write 1 again to store a new angle, 0 to switch it off.

4. Setup

· Autopilot Parameters

Set these on the autopilot and reboot it.

Parameter	Value	Description
CAN_P1_DRIVER	1	Enable the CAN1 port with driver 1 (reboot after the change)
CAN_P1_BITRATE	1000000	1 Mbit/s (default)
CAN_D1_PROTOCOL	1	DroneCAN
CAN_D1_UC_ESC_BM	15	Outputs sent to DroneCAN ESCs: 15 = outputs 1–4 (motors 1–4). 0 = no CAN throttle at all
CAN_D1_UC_ESC_OF	0	Outputs skipped before ESC 0. Motors on outputs 5–8: CAN_D1_UC_ESC_BM 240 and CAN_D1_UC_ESC_OF 4

4. Setup

· ESC Parameters (1)

In Mission Planner: SETUP → Optional Hardware → DroneCAN/UAVCAN → MAVlink-CAN1 → Start, then Menu → Parameters on the ESC (AE14S_60A_V1.40). Change the value and Write.

Parameter	Value	Description
esc_index	0 ... 19	Position in the CAN throttle command: 0 = first marked output, 1 = the next ... Every ESC needs its own number (default 0)
node_id	0 / 1 ... 125	0 = node number from the autopilot (dynamic allocation, default); 1–125 = fixed number
dir	0 / 1	Spin direction. Change it here, never by swapping motor leads
prop_align	0 / 1	1 = store the present propeller angle and hold it at 0 % throttle; 1 again = new angle; 0 = off. Motor stopped, after a passed auto_cal
auto_cal	1 / 2	1 = measure the motor and calibrate the encoder (about 20 s, slow turn, propeller off, disarmed, cold motor). 2 = forget the measured values (defaults from the next power-up). Reads 0
can_pwm	0 / 1	Throttle source: 0 = CAN (default), 1 = PWM input (option, see the pinout)

4. Setup

· ESC Parameters (2)

Parameter	Value	Description
motor_rs_mohm	read only	Phase resistance measured by auto_cal (mΩ, 0 = none)
motor_ls_uh	read only	Inductance measured by auto_cal (µH, 0 = none)
motor_flux_vphz	read only	Flux measured by auto_cal (V/Hz, 0 = none)
motor_meas	read only	Result of the last auto_cal: 1 = saved; 11–23 = not saved, the reason is listed in the user guide
motor_pole_pairs	read only	Pole pairs found by auto_cal (14 for AM-6212 / AM-6215, 0 = none)

5. auto_cal Result

· motor_meas

After auto_cal the ESC shows the result in motor_meas.

Code	Meaning
0	Not measured since this power-up
1	Measured values saved
11	Failed: resistance
12	Failed: inductance
13	Failed: flux
14	Values out of range — not saved
15	The motor was turning — not measured
16	The rotor did not follow — not saved (check the propeller and for binding)
17	Stopped: the aircraft was armed
18	After an in-air restart — power-cycle, then try again
19	The motor may be hot from running — let it cool, then try again
20	The ESC is hot — let it cool, then try again
21	Autopilot disarm not confirmed — not started
22	The two passes did not agree — the previous values are kept
23	The motor chip did not answer

6. Firmware

· Update over DroneCAN

Firmware is updated over DroneCAN through the autopilot — the case stays closed. Current release: App V1.40 with DroneCAN 1.13; the DroneCAN node name shows the app version (AE14S_60A_V1.40). Update files are supplied by novaX.

An update takes about 45 seconds. Keep the battery connected until the ESC restarts and its node name shows the new version.

7. Safety

· Before You Fly

- Check the battery polarity before connecting — a reversed battery destroys the ESC at once.
- Stay within 6–14S LiPo (18–60 V).
- Remove the propeller for auto_cal and whenever you work near the motor; fit it only to set the parking angle.
- Do not swap motor leads after auto_cal — change the direction with dir.
- Keep the battery leads short: long leads raise the voltage spikes the ESC has to take.
- Mount the ESC in the airflow. Heat lowers its current limit (down to 50 %), it does not stop the motor.
- Settings and firmware updates are refused while the motor turns or the autopilot is armed.

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